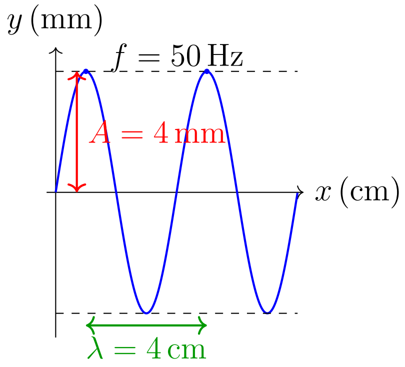
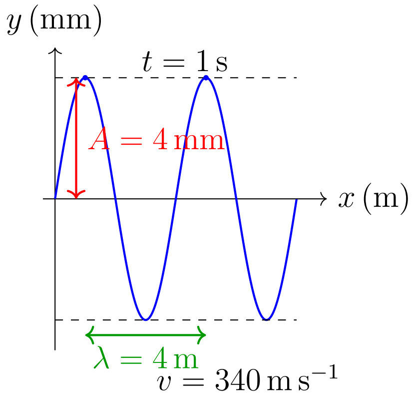
(KTXFI1EBNF) Physics I. Homework No. 5

May 26, 2026

1. The wave shown in the figure was generated by a 50 Hz resonator. Determine:
   a) amplitude of the wave
   b) frequency of the wave
   c) wavelength of the wave
   d) propagation speed of the wave
   e) period of the wave



2. A wave propagating along a string can be defined as follows (with x in meters, t in seconds): $y(x,t) = 0.03\,\mathrm{m}\sin(50\,\mathrm{s}^{-1}t - 5\,\mathrm{m}^{-1}\,x)$. Determine the:
   a) amplitude
   b) frequency
   c) wavelength
   d) speed
3. Determine the amplitude, frequency, and wavelength of the wave shown in the figure below, if the propagation speed of the wave is 340 m/s. Write a solution for the wave if at t=1 s the wave appears as shown in the figure.



4. For a wave propagating along a string with a frequency of 10 Hz, $\lambda = 80\,\mathrm{cm}$, A=3 mm, write the solution in SI units.
5. A wave solution is given by $y = 0.3\sin(15x - 600t)\,mm$, where x is in meters and t is in seconds. For the transverse wave at x=30 cm, determine:
   a) its speed
   b) its acceleration
   c) What is the displacement of the point at t=4 s?

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